100 Days of Linux - Day 024

Title: Using Wildcards (*, ?, and [])

Today's Goal
Learn how to use Linux wildcards to match multiple files and patterns efficiently.

Why This Matters
Wildcards save time when working with many files. Developers and data engineers use them with commands like ls, cp, mv, rm, and find.

Command of the Day: Wildcards

***** = Matches zero or more characters
Example: **ls *.txt**

? = Matches exactly one character
Example: **ls file?.txt**

[] = Matches one character from a set or range
Example: **ls report[1-3].txt**

Explanation: Wildcards help you select groups of files without typing every filename.

Beginner Lesson
Instead of specifying every filename manually, you can use wildcard patterns. For example, '*.txt' matches all text files, while 'file?.txt' matches file1.txt but not file10.txt.

Practice Questions

1. Create files named report1.txt, report2.txt, report3.txt, and notes.txt. List only the .txt files using a wildcard.

2. Create files named file1.txt, file2.txt, file10.txt, and use a wildcard to display only file1.txt and file2.txt.

3. Create files named testA.txt, testB.txt, and testC.txt. Use square brackets to list only testA.txt and testB.txt.

4. Copy all .txt files into a directory named backup using a wildcard.

5. Display your current working directory and list the contents of the backup directory.

Hints

Remember: * matches many characters, ? matches exactly one character, and [] matches one character from a set.

Mini Challenge

Create five CSV files and three TXT files. Use wildcards to list only CSV files, then copy only TXT files into a folder named archive.

Common Mistakes
Confusing '*' with '?'. The asterisk matches any number of characters, while '?' matches exactly one.

Did You Know?
Wildcards are expanded by the shell before the command runs, which is why so many Linux commands support them.

Pro Tip
Combine wildcards with commands like cp, mv, rm, and grep to work with many files quickly.

Answer Key (Instructor Only)
Q1: ls *.txt
Q2: ls file?.txt
Q3: ls test[AB].txt
Q4: mkdir backup && cp *.txt backup/
Q5: pwd && ls backup

Homework
Create 10 practice files with different names and experiment with *, ?, and [] until you understand the matching patterns.

Next Day Preview
Tomorrow is Day 25, your first Mini Project, combining file management, searching, and wildcards into one practical exercise.